Special Optimal Thermal Cell: Correlator-Designed Supercells for Effective-Harmonic Phonons and Thermodynamics

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Finite-temperature effective-harmonic calculations commonly estimate thermal displacement statistics from many stochastically sampled configurations. We introduce the Special Optimal Thermal Cell (SOTC), which instead designs configurations to reproduce selected statistics. Given a harmonic trial model, SOTC constructs collective displacements in a fixed periodic supercell so that shell-averaged displacement correlations match thermal targets within the force-constant interaction range. First-principles forces from these structures are fitted to a symmetry-constrained effective second-order force-constant (FC2) model. The fitted phonons then provide new targets for the next iteration. The method adapts the special-structure approach used for chemical disorder to thermal motion: ensemble pair correlations are imposed through configuration design rather than recovered only by stochastic averaging. Configuration design remains separate from force evaluation and can be paired with different electronic-structure engines. Applications to metal and rocksalt systems demonstrate fixed-volume phonon and thermodynamic calculations, including non-analytic LO–TO corrections for polar crystals. Matched stochastic controls give comparable accuracy at equal force-call counts and expose sensitivity to initialization. SOTC is a deterministic configuration-design method for effective-harmonic inference.

I. INTRODUCTION

High-temperature phonon properties (renormalized dispersions, soft-mode instabilities, thermal expansion, heat capacity) encode the anharmonic response of a crystal to atomic vibrations with amplitudes reaching several percent of the lattice constant. Accurate first-principles descriptions of these properties are essential for predicting phase stability [1], thermal conductivity [2], electron–phonon coupling [3], and thermodynamic databases for materials design [4].

Finite-temperature lattice dynamics can be formulated at several distinct levels. The temperature-dependent effective potential framework fits effective second- and higher-order IFCs to finite-temperature force data [5–8]. The self-consistent harmonic approximation (SCHA) is the variational effective-harmonic level; the stochastic SCHA (SSCHA) is a stochastic sampling and optimization implementation of that variational theory [9–11]. At the effective-harmonic level, stochastic TDEP and SSCHA can reach the same stationary solution and free energy when they use consistent ensembles, estimators, and converged sampling; their numerical minimization procedures differ. Mode-coupling theory provides a separate non-perturbative description of anharmonic lattice dynamics [12], while ab initio molecular dynamics samples classical nuclear trajectories on the chosen electronic potential-energy surface [13]. These approaches do not target identical observables or approximations in general; meaningful cost or accuracy comparisons require matched cells, electronic settings, temperatures, and physical outputs.

In a parallel development, Zunger and coworkers [14] demonstrated that a random substitutional alloy can be mimicked statistically by a single small supercell, the Special Quasirandom Structure (SQS), whose chemical correlation functions match those of the ideal random alloy up to a spatial cutoff. The key observation is that structure encodes information: the correct spatial correlations are imposed by construction rather than recovered by averaging over an ensemble.

We apply the special-structure idea to displacement design. A harmonic trial model defines displacement–displacement correlations $C_{ab}^{\alpha\beta}(\mathbf{R}, T)$. SOTC searches a fixed periodic supercell for a collective displacement pattern whose selected shell-averaged correlators approximate these targets up to the IFC cutoff. DFT forces on the designed configuration are then used in a symmetry-constrained ridge fit of second-order IFCs, and the fitted phonons update the target correlators. The small iteration counts reported below are empirical observations for the present benchmarks, not a proof that a single configuration identifies an arbitrary IFC tensor.

SOTC is thus a deterministic configuration-design and effective-FC2 fitting procedure. It does not minimize an anharmonic variational free energy or generate a time trajectory. Its proposed distinction is the explicit optimization of prescribed spatial correlator moments before the force calculation, rather than a general replacement for ensemble or trajectory methods.

The most direct applications are reproducible effective-FC2 construction for dynamically stable, nonmagnetic crystals and fixed-volume phonon-DOS thermodynamics, including C_V, ZPE, F_{vib}, and S_{vib}. The deterministic FC2 model can also serve as a harmonic reference for a larger finite-temperature workflow. When a suffi-

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ciently large ensemble of forces is available, the independently tested constrained regression route can be used to extract FC3 and FC4. The demonstrated application domain does not include strongly non-Gaussian phases, anharmonic free energies, phonon lifetimes, or thermal conductivity.

This paper is organized as follows. Section II presents the thermal-displacement correlators, SOTC objective, IFC extraction, phonon reconstruction, and self-consistency loop. Section III describes the QE computational settings. Section IV reports the effective-FC2 benchmarks, same-volume three-material production-refit non-analytic corrections, an independent NaCl control, and the independent FC2/FC3/FC4 validation. Section V discusses the validated scope, limitations, computational cost, and related anharmonic methods. Section VI concludes.

II. THEORY

A. Thermal Displacement Correlators

Consider a crystal with n_b basis atoms and $N_c = N_1N_2N_3$ primitive cells, so that the supercell contains n_bN_c atoms. Let $u_a^\alpha(\mathbf{R})$ be the displacement of basis atom a in cell $\mathbf{R}$. With the Fourier convention defined in Appendix A, the equal-time canonical correlator of a harmonic lattice is [15]:

$$C_{ab}^{\alpha\beta}(\mathbf{R},T) = \langle u_a^\alpha(\mathbf{0})\, u_b^\beta(\mathbf{R})\rangle_T = \frac{\hbar}{2N_c\sqrt{M_aM_b}} \sum_{\mathbf{q},s} \frac{e_a^{\alpha*}(\mathbf{q}s)\, e_b^\beta(\mathbf{q}s)}{\omega_{\mathbf{q}s}} \coth\left(\frac{\hbar\omega_{\mathbf{q}s}}{2k_BT}\right) e^{-i\mathbf{q}\cdot\mathbf{R}}. \tag{1}$$

Here $\mathbf{q}$ runs over the N_c wave vectors commensurate with the Born–von Kármán cell, s labels a phonon branch, and the polarization vectors obey the normalization stated in Appendix A. The remaining symbols have their usual meanings. Equation (1) is standard harmonic-lattice theory; a derivation and the phase convention are given in Appendix A.

The coth factor encodes the full quantum Bose–Einstein statistics, smoothly interpolating between the zero-point limit ($T \to 0$: coth $\to 1$) and the classical equipartition limit ($k_BT \gg \hbar\omega$: coth $\to 2k_BT/\hbar\omega$).

To reduce the tensor correlator to the scalar target used in the present work, let $\mathcal{S}_R^{ab}$ denote the set of lattice vectors belonging to the crystallographic shell of radius R for basis pair (a,b), with multiplicity N_R^{ab}. The shell and Cartesian trace average is defined by

$$\bar{C}_2^{ab}(R,T) = \frac{1}{3N_R^{ab}} \sum_{\mathbf{R}'\in\mathcal{S}_R^{ab}} \sum_\alpha C_{ab}^{\alpha\alpha}(\mathbf{R}',T). \tag{2}$$

Thus R labels a discrete neighbor shell; it is not a continuum spherical average. In the reported implementation, pair distances are rounded to finite 0.05 Å radial bins, separately for every ordered basis/species-pair channel. This numerical grouping approximates the ideal crystallographic shells in Eq. (2); it is not a full space-group orbit construction. The tolerance is now an explicit archived input. This scalar reduction also discards anisotropic and off-diagonal correlation information, a limitation discussed in Sec. V.D.

For a monatomic cubic crystal, the on-site value is the one-component isotropic displacement parameter $U_{\rm iso} = \frac{1}{3}\langle|\mathbf{u}|^2\rangle$:

$$U_{\rm iso}(T) = \bar{C}_2(0,T) = \frac{\hbar}{6N_cM} \sum_{\mathbf{q},s} \frac{1}{\omega_{\mathbf{q}s}} \coth\left(\frac{\hbar\omega_{\mathbf{q}s}}{2k_BT}\right). \tag{3}$$

Equation (3) follows from the trace of Eq. (1) and the polarization-vector completeness relation. The total three-dimensional MSD is $3U_{\rm iso}$, while crystallographic Debye–Waller factors use $B = 8\pi^2U_{\rm iso}$. For compounds, species-resolved displacements are evaluated from the full eigenvector-weighted mode sum rather than a species-averaged mass and the total DOS. The displacement norm used to design a SOTC configuration is documented separately from this one-component quantity to avoid a factor-of-three ambiguity.

B. The SOTC Objective Function

Let Ω be a candidate supercell of n atoms (with $n \ll N$) and let $\{\mathbf{u}_i\}$ be a set of atomic displacements assigned to those atoms. The **SOTC objective function** measures the mismatch between the selected correlators of a finite displacement vector and the harmonic trial targets:

$$\mathcal{Q}(\mathbf{u}) = \sum_{a,b} \sum_{R<r_{\rm cut}} w_{ab}(R) \left[\bar{C}_{2,ab}^{\rm cell}(R;\mathbf{u}) - \bar{C}_{2,ab}(R,T)\right]^2. \tag{4}$$

Here $\bar{C}_{2,ab}^{\rm cell}$ is formed with the same shell, trace, and species averaging as Eq. (2), and $r_{\rm cut}$ is the real-space IFC cutoff. Equation (4) is introduced in this work; its gradient is given in Appendix C. A positive distance weight may be chosen, for example

$$w_{ab}(R;\xi_w) = \exp(-R/\xi_w), \tag{5}$$

where ξ_w is a numerical weighting length whose adequacy must be tested by cutoff and dispersion convergence. We do not identify ξ_w with a universal phonon mean free path: a finite dynamical phonon lifetime does not, by itself, prove exponential decay of an equal-time displacement correlator. No lifetime-based exponential decay or universal T^{-1}/T^{-3} scaling law is assumed in this formulation.

Minimization of $\mathcal{Q}$ over the displacement field $\{\mathbf{u}_i\}$ subject to $|\mathbf{u}_i| \le u_{\rm rms}(T)$ is performed by gradient descent with random restarts. The initial guess

is the harmonic displacement pattern obtained by sampling each phonon mode with amplitude $A^{(0)}_{\mathbf{q}s} = \sqrt{\hbar \coth(\hbar\omega_{\mathbf{q}s}/2k_BT)/2\omega_{\mathbf{q}s}}$. The optimization adjusts mode amplitudes and phase relations to minimize the correlator mismatch while keeping displacements physically bounded.

C. IFC Extraction by Ridge Regression

Let $F_i^\alpha(\mathbf{u})$ and $F_i^\alpha(\mathbf{0})$ be forces on the displaced and reference structures at the same cell. The response force $\Delta F_i^\alpha = F_i^\alpha(\mathbf{u}) - F_i^\alpha(\mathbf{0})$ removes any residual reference force. To harmonic order, the collective-displacement relation with translational invariance written explicitly is

$$\Delta F_i^\alpha = -\sum_{j,\beta} \Phi_{ij}^{\alpha\beta}\,(u_j^\beta - u_i^\beta) + \mathcal{O}(u^2). \tag{6}$$

Equation (6) is the standard harmonic force–displacement relation [15]; the reference subtraction and constrained matrix assembly used here are detailed in Appendix D. This constitutes an underdetermined system when the number of IFC parameters N_{IFC} exceeds $3n$ (the dimension of the force vector). We use Tikhonov (ridge) regression to stabilize the inversion:

$$\hat{\mathbf{\Phi}} = -\left(\mathbf{G}^T\mathbf{G} + \lambda\mathbf{I}\right)^{-1}\mathbf{G}^T\Delta\mathbf{F}. \tag{7}$$

The geometry matrix $\mathbf{G}$ contains the relative displacements in Eq. (6), $\Delta\mathbf{F}$ stacks the response forces, and λ is the ridge parameter. Equation (7) is standard Tikhonov regularization applied to the symmetry-reduced force model; the constrained construction is detailed in Appendix D. The acoustic sum rule is imposed as a hard linear constraint, and an optional central-force projection further reduces the parameter space. These constraints, the finite cutoff, and ridge regularization are physical and statistical priors, and their influence must be distinguished from information provided by the DFT forces.

For one n-atom configuration, $\mathbf{G}$ has $3n$ rows and hence algebraic rank no larger than $3n$. Full identification requires the rank to be compared with the number of independent IFC parameters after all symmetry and sum-rule reductions. Ridge regularization stabilizes a rank-deficient fit but does not create missing information. The validation therefore reports parameter counts, singular-value tolerance, rank, and condition number for each actual matrix rather than asserting coverage from the iteration count alone.

D. Phonon Reconstruction

From the extracted IFCs $\hat{\Phi}_{ij}^{\alpha\beta}(\mathbf{R})$, the dynamical matrix is constructed as:

$$D_{ij}^{\alpha\beta}(\mathbf{q}) = \frac{1}{\sqrt{M_i M_j}} \sum_{\mathbf{R}} \hat{\Phi}_{ij}^{\alpha\beta}(\mathbf{R})\, e^{i\mathbf{q}\cdot\mathbf{R}} \tag{8}$$

Equation (8) is the standard Fourier construction of the harmonic dynamical matrix from real-space IFCs [15]; the retained real-space cutoff and non-analytic polar correction determine its domain of validity.

Phonon dispersion: the $3n_b$ branches $\omega_{\mathbf{q}s}$ follow from eigenvalues of $D(\mathbf{q})$, evaluated on a dense Monkhorst–Pack grid. The phonon density of states (DOS) is obtained by linear tetrahedron integration with Gaussian broadening $\sigma = 0.1$ THz.

Thermodynamic properties follow directly from the phonon DOS $g(\omega)$:

$$C_V(T) = k_B \int_0^{\omega_{\max}} g(\omega) \left(\frac{\hbar\omega}{2k_BT}\right)^2 \sinh^{-2}\left(\frac{\hbar\omega}{2k_BT}\right) d\omega \tag{9}$$

$$F_{\mathrm{vib}}(T) = k_BT \int_0^{\omega_{\max}} g(\omega) \ln\left[2\sinh\left(\frac{\hbar\omega}{2k_BT}\right)\right] d\omega \tag{10}$$

$$\mathrm{ZPE} = \frac{\hbar}{2}\int_0^{\omega_{\max}} g(\omega)\,\omega\, d\omega \tag{11}$$

$$U_{\mathrm{iso}}(T) = \frac{\hbar}{6M}\int_0^{\omega_{\max}} \frac{g(\omega)}{\omega} \coth\left(\frac{\hbar\omega}{2k_BT}\right) d\omega, \qquad \int_0^{\omega_{\max}} g(\omega) d\omega = 3. \tag{12}$$

Equation (12) is the monatomic one-component displacement obtained from Eq. (3) when the total DOS is normalized to three modes per atom. For a compound, species-resolved eigenvector weights and masses are required; a species-averaged mass multiplying the total DOS is not used. The harmonic thermodynamic identities in Eqs. (9)–(12) are standard consequences of the independent-phonon partition function [15]. The spectral Debye temperature is the second-moment Debye temperature $T_D^{\mathrm{spec}} = (\hbar/k_B)\sqrt{5\langle\omega^2\rangle_g/3}$, where $\langle\omega^2\rangle_g = \int g(\omega)\,\omega^2\, d\omega / \int g(\omega)\, d\omega$ is the mean-square phonon frequency over the DOS; this uses the Debye relation $\langle\omega^2\rangle = 3\omega_D^2/5$. The caloric Debye temperature $T_D^{\mathrm{cal}}(T)$ is defined by inversion of the Debye model for $C_V(T)$.

E. Self-Consistent Loop

The SOTC method is closed into a self-consistency cycle:

1. **Initialize**: start from a Debye trial phonon model $\omega^{(0)}(\mathbf{q})$.
2. **Compute target correlators**: $\bar{C}_2(R,T)$ from Eqs. (1)–(2) using current phonon model.
3. **Design configurations**: with the supercell fixed, minimize $\mathcal{Q}(\mathbf{u})$ [Eq. (4)] to generate N_{ens} displacement configurations.
4. **DFT forces**: evaluate the undisplaced reference and all N_{ens} displaced configurations.
5. **Extract IFCs**: ridge regression from Eq. (6).
6. **Update phonon model**: diagonalize $D(\mathbf{q})$ from new IFCs.
7. **Check convergence**: the archived benchmarks use $\|\Delta\bar{C}_2\|_2 < 3 \times 10^{-3}$ Å^2, together with the material-specific minimum iteration counts reported in the calculation inputs. If not converged, return to step 2.

The Debye model supplies only the first target in the intended self-consistent workflow; subsequent targets depend on the fitted material-specific IFCs. Nevertheless, finite-data priors and numerical safeguards can retain initialization dependence, which must be measured rather than assuming a unique fixed point. The matched control below brackets the nominal Debye temperature by ±25%.

Each iteration evaluates N_{ens} displaced configurations and an undisplaced reference configuration. In the archived workflow the reference is recomputed once per iteration; hence the force-call count is $N_{\text{iter}}(N_{\text{ens}}+1)$, not N_{iter}. The archived benchmarks reach their stopping criterion in 2–5 iterations with $N_{\text{ens}} = 1$–8, as detailed in Table I. Iteration count is therefore not used as a proxy for DFT cost.

III. COMPUTATIONAL DETAILS

A. DFT Calculations

All force calculations used Quantum ESPRESSO 7.5 [16], PBE [17], and pseudopotentials from the SSSP efficiency library [18]. The selected potentials were ONCV for Ag, Au, and Mo; PAW for Al, Cu, Mg, O, and Pb; and ultrasoft for Na, Cl, Te, and W. Wave-function cutoffs (Ry) were Al 40, Ag 50, Au 45, Cu 45, Mo 40, W 30, NaCl 60, MgO 80, and PbTe 44. Charge-density cutoffs were four times the wave-function cutoff for ONCV and eight times it for PAW/ultrasoft potentials. FCC and BCC supercells used a Gamma-centered $2 \times 2 \times 2$ k mesh; rocksalt supercells used Gamma only. Metals used Marzari–Vanderbilt cold smearing [19] of 0.02 Ry, while the insulators used fixed occupations. Software and file-level provenance is given in Appendix J.

Experimental lattice constants and ideal high-symmetry fractional coordinates were used for the archived benchmarks. An undisplaced single-point calculation at the same cell defines the reference force and stress. IFC fits use $\mathbf{\Delta F} = \mathbf{F}(\mathbf{u}) - \mathbf{F}(\mathbf{0})$, as in Eq. (6), rather than assuming $\mathbf{F}(\mathbf{0}) = 0$. In the independent QE campaigns, the reference-force RMS values are 1.67×10^{-5}, 2.45×10^{-5}, and 8.34×10^{-7} eV Å^{-1} per Cartesian component for Al, W, and NaCl, respectively, compared with induced thermal-force RMS values of 0.395, 1.523, and 0.598 eV Å^{-1}. The induced-to-reference RMS ratios are 2.37×10^4, 6.22×10^4, and 7.17×10^5, and the maximum reference-force atom norms are 4.19×10^{-5}, 5.83×10^{-5}, and 4.11×10^{-6} eV Å^{-1}. The corresponding hydrostatic stresses are 0.487, −0.761, and 0.322 GPa. Thus the ideal coordinates have negligible residual atomic force despite finite cell stress. Sensitivity to PBE-relaxed lattice constants is not quantified.

B. SOTC Supercell Setup

The supercell geometry (Hermite Normal Form matrix $\mathbf{H}$) is selected once before the self-consistency loop by enumerating all HNF candidates whose number of primitive cells n satisfies $\max(2, n_{\text{sc}} - 2) \le n \le n_{\text{sc}}$, where n_{sc} is the user-specified target. Candidates are ranked by a two-criterion score: nearest-neighbor shell completeness first, then supercell lattice condition number (a measure of cell cubicity); the best candidate at or below the target size is adopted. This upper bound ensures the supercell never exceeds the intended size.

FCC metals (Al, Ag, Au, Cu) used a $2 \times 2 \times 2$ conventional-cell supercell (4 atoms/cell × 8 = 32 atoms); BCC metals (Mo, W) used a 64-atom supercell (2 atoms/cell × 32 conventional cells) and rocksalt compounds (NaCl, MgO, PbTe) used a $2 \times 2 \times 2$ conventional-cell supercell (8 atoms/cell × 8 = 64 atoms).

The experimental lattice parameters are provided in the archived inputs. Table I reports the IFC cutoffs and force-call accounting. The SOTC objective groups pair distances into 0.05 Å radial bins and keeps ordered basis/species-pair channels separate; this shell-bin width is now an explicit calculation input. The ridge coefficient was 10^{-3} in the reported convention, and a central-force bond projection was applied to all primary benchmarks. Because sensitivity to that physical prior is not quantified, the reported FC2 results are conditional on the constrained fit.

The archived runs used material-specific ensembles of $N_{\text{ens}} = 1$–8 configurations per iteration, as listed in Table I. These force rows do not by themselves establish full IFC coverage. Complete symmetry-reduced matrix dimensions, numerical ranks, singular-value thresholds, and condition numbers are unavailable for every primary benchmark. Scalar self-consistency and FC2 identifiability are therefore treated as separate diagnostics.

C. Phonon Post-Processing

Phonon DOS calculations used a $30 \times 30 \times 30$ Monkhorst–Pack mesh (27,000 q points), while band dispersions were interpolated from the real-space IFCs along the paths stored with the released band data: Γ-X-W-K-Γ-L for FCC and rocksalt and Γ-H-N-Γ-P-H for BCC. Thermal properties were computed on an $8 \times 8 \times 8$ grid (512 q-points) for $C_V(T)$, $F_{\mathrm{vib}}(T)$, and $S_{\mathrm{vib}}(T)$. Gaussian broadening $\sigma = 0.1$ THz was applied to the phonon DOS. Convergence of C_V with q-mesh was checked by comparing $8 \times 8 \times 8$ and $16 \times 16 \times 16$ grids for Al and MgO; the difference is < 0.05 J/(mol K) in both cases. This establishes numerical stability of that integral for the two tested materials, not universal q-mesh convergence for every system or branch-resolved observable.

D. Experimental Reference Data

Experimental Debye temperatures and heat capacities were taken from the NIST-JANAF tables [20] and Kittel [21]. References for individual systems: Al [22], Ag [23], Au [24], Cu [25, 26], Mo [27], W [28], NaCl [29], MgO [30, 31], PbTe [32, 33]. For comparison of phonon dispersions, we use neutron scattering data from: Cu (Ref. [26]), Al (Ref. [34]), NaCl (Ref. [35]).

IV. RESULTS

A. Self-Consistency Convergence

The Debye-temperature comparison contains the SOTC FC2 and experimental values. Available external-method series lack calculation-level provenance and matched settings and are therefore outside this comparison.

The archived runs stop after 2–5 self-consistency iterations under the material-specific minimum-iteration rule and the correlator threshold stated in Sec. II.E. This quantity measures convergence of the selected scalar metric; full-dispersion convergence is a separate, unmeasured diagnostic in the primary archive.

Table I shows that the present runs themselves use ensembles and require 2–32 displaced calculations, or 4–36 force calculations when the repeated reference evaluations are counted. No external-method cost-reduction factor is inferred from these numbers because no comparison dataset was generated with a matched cell, electronic-structure setup, sampling protocol, IFC rank, and dispersion accuracy target.

B. Phonon Dispersions and DOS

Figure 1 displays the SOTC effective-FC2 dispersions and DOS. The curves expose branch topology and instabilities but do not include explicit FC3 or FC4 self-energies. Figure 1 is interpreted descriptively because its Cu and Al experimental curves are not available as machine-readable point arrays and its production temperatures are generally unmatched. The recoverable tabulated NaCl neutron points are instead shown in Fig. 4. Direct pointwise sorted-mode comparisons to independent conventional QE finite-displacement references are reported in Fig. 2 for Al, W, and NaCl.

The cited literature comprises a 293 K Cu dispersion [26], Al dispersions at 80 and 300 K [34], and tabulated 80 K NaCl neutron data [35]. The 39 NaCl points from Table I of Ref. [35] along Γ–X and Γ–L are overlaid in Fig. 4; their reported angular frequencies were converted by $\nu = \omega/(2\pi)$. The Cu and Al sources provide plotted curves rather than machine-readable tables, so they are not digitized or assigned a numerical error here. Only the 300 K Al literature curve is temperature matched to a production panel; the 1200 K Cu and 700 K NaCl production panels remain qualitative temperature-mismatched comparisons. The NaCl zone-center optic comparison is quantified separately below. The Au imaginary modes make a stable-branch comparison indeterminate with respect to cell, cutoff, regularization, symmetry projection, electronic convergence, and iteration.

Figure 2 compares the stored SOTC bands directly with independent conventional finite-displacement QE FC2 references at the same PBE lattice parameter and pseudopotential. Pointwise sorted-mode frequency MAE, RMSE, and maximum absolute error are 0.377, 0.493, and 1.200 THz for Al; 0.242, 0.298, and 0.731 THz for W; and 0.700, 0.931, and 1.786 THz for NaCl. Neither the SOTC nor reference paths contain modes below -0.05 THz. The reference campaigns use $3 \times 3 \times 3$ primitive cells and the specified electronic meshes reported below, so this is a cross-archive physical-output comparison rather than a strictly cell-matched sampling test.

The same figure provides a random/canonical-Gaussian baseline by repeatedly fitting FC2 to ten independently chosen subsets of the 100-snapshot QE campaigns. These refits use the primary SOTC cutoff, ridge coefficient, central-force projection, and acoustic-sum-rule construction. At the same numbers of displaced configurations as the primary SOTC archives (10 for Al, 12 for W, and 15 for NaCl), the median random-subset dispersion RMSEs are 0.397, 0.203, and 0.560 THz, respectively, compared with 0.493, 0.298, and 0.931 THz for SOTC. The canonical subsets have lower median errors for all three materials. The primary and control cells, displacement ensembles, and electronic meshes still differ, so these ratios are not a matched speedup.

To isolate displacement selection, a second control uses exactly the same independent $3 \times 3 \times 3$ primitive-cell campaigns, electronic settings, reference-force subtraction, drift projection, IFC cutoff, ridge coefficient, central-force projection, acoustic sum rule, 20 held-out snap-

TABLE I. Archived SOTC iteration and force-call accounting. N_{iter} is the number of self-consistency iterations, N_{ens} the displaced configurations evaluated per iteration, $N_{\mathrm{disp}} = N_{\mathrm{iter}} N_{\mathrm{ens}}$, and $N_{\mathrm{force}} = N_{\mathrm{iter}}(N_{\mathrm{ens}}+1)$ includes the undisplaced reference recomputed by the archived workflow. The counts quantify workflow cost independently of IFC or dispersion convergence.

Material	T_{des} (K)	N_{atoms}	r_{cut} (Å)	N_{iter}	N_{ens}	N_{disp}	N_{ref}	N_{force}
Al	300	32	4.5	2	5	10	2	12
Ag	1000	32	4.5	4	5	20	4	24
Au	1200	32	4.5	2	1	2	2	4
Cu	1200	32	4.5	4	8	32	4	36
Mo	1500	64	5.0	2	1	2	2	4
W	1500	64	5.0	4	3	12	4	16
NaCl	700	64	4.3	5	3	15	5	20
MgO	1500	64	3.2	4	3	12	4	16
PbTe	700	64	5.0	4	5	20	4	24

shots, and reciprocal paths for both methods. At $N = 1, 2, 4, 8, 16$ displaced-force calls, 20 seeded canonical-Gaussian permutations of the same 80-snapshot training pool are compared with three independently optimized correlator-matched sequences initialized at $0.75T_D$, T_D, and $1.25T_D$. This is a controlled first-stage displacement-selection ablation, not a rerun of the complete self-consistent outer loop; the initialization target and optimizer start vary together.

Figure 3 reports both pointwise sorted-mode dispersion RMSE and held-out force RMSE. At 16 calls, the canonical dispersion medians are 0.412, 0.207, and 0.564 THz for Al, W, and NaCl. The nominal-T_D designs give 0.425, 0.223, and 0.548 THz, while the best of the three initializations gives 0.404, 0.178, and 0.459 THz. Conversely, the $0.75T_D$ designs give 0.574, 0.377, and 0.928 THz. The corresponding canonical held-out force-RMSE medians are 0.104, 0.281, and 0.332 eV Å^{-1}; the nominal designs give 0.104, 0.282, and 0.333 eV Å^{-1}. Thus the favorable initialization can outperform the canonical median for the tested dispersion metric, but the gain is material- and initialization-dependent rather than universal.

For NaCl, the non-analytic polar correction was evaluated explicitly. Primitive-cell QE 7.5 DFPT calculations used PBE, the same Na and Cl pseudopotentials as the force calculations, 60/480 Ry wave-function/charge cutoffs, an $8 \times 8 \times 8$ electronic mesh, 10^{-10} Ry SCF convergence, and 10^{-14} DFPT convergence. The charge-neutrality-projected tensors are $Z^*_{\mathrm{Na}} = +1.110915\,\mathbf{I}$, $Z^*_{\mathrm{Cl}} = -1.110915\,\mathbf{I}$, and $\varepsilon_\infty = 2.470969\,\mathbf{I}$. Applying these tensors through the Phonopy finite-displacement and lattice-dynamics software package [36] to the matched central finite-displacement QE FC2 force constants changes the directional Γ-point optic limit from 4.369 (TO) to 7.264 THz (LO), a 2.895 THz splitting (Fig. 4).

Archival infrared dielectric data in NBS Special Publication 301 report zone-center TO and LO limits of 164 and 268 cm^{-1}, respectively [37]. These correspond to 4.917 and 8.034 THz and are overlaid in Fig. 4. Thus the present QE harmonic calculation is lower by 0.547 THz (11.1%) for TO and 0.770 THz (9.6%) for LO; its splitting is 0.223 THz (7.2%) below the 3.118 THz optical value. The calculated LO/TO ratio, 1.663, is 1.7% above the optical ratio of 1.634. The same figure also shows the 39 tabulated 80 K neutron points along Γ–X and Γ–L [35]. Because those points are temperature mismatched to the fixed-cell calculation and do not cover the entire plotted path, they are shown without a global RMSE. Room-temperature IXS and SSCHA calculations further show that the NaCl LO spectral function can contain multiple peaks along Γ–L [38]; the harmonic-NAC curve is not interpreted as a finite-temperature spectral function.

For a production-model test of all three polar compounds, primitive-cell DFPT was evaluated at the corresponding production lattice parameters $a = 5.73$, 4.26, and 6.51 Å for NaCl, MgO, and PbTe, using the same 60/480 Ry cutoffs, $8 \times 8 \times 8$ mesh, and convergence thresholds. The charge-neutrality-projected cubic tensor pairs $(Z^*_{\mathrm{cation}}, \varepsilon_\infty)$ are $(1.110915, 2.470969)$, $(1.973385, 3.167143)$, and $(5.766700, 28.826866)$, respectively, with opposite anion charges.

These tensors were applied through Phonopy's Wang non-analytic formulation to FC2 refits of the actual selected production iterations: iterations 5, 4, and 2, containing 3, 3, and 5 displaced configurations for NaCl, MgO, and PbTe. The refits use each iteration's repeated equilibrium QE snapshot for reference subtraction. Directional Γ-point TO/LO/splitting values are 6.223/8.509/2.286 THz, 14.097/22.144/8.047 THz, and 2.903/4.219/1.315 THz, respectively. The converted no-NAC force constants agree with the native SOTC lattice-dynamics implementation to better than 2×10^{-6} THz RMSE. Because the selected in-memory IFC objects were not serialized, these reproducible snapshot refits are not asserted to be identical to the immutable no-NAC arrays in Fig. 1. The PbTe result is scalar-relativistic and does not test spin–orbit sensitivity.

C. Heat Capacities

Table II reports the effective-harmonic C_V values obtained by integrating the SOTC DOS through Eq. (9), together with calorimetric constant-pressure values from

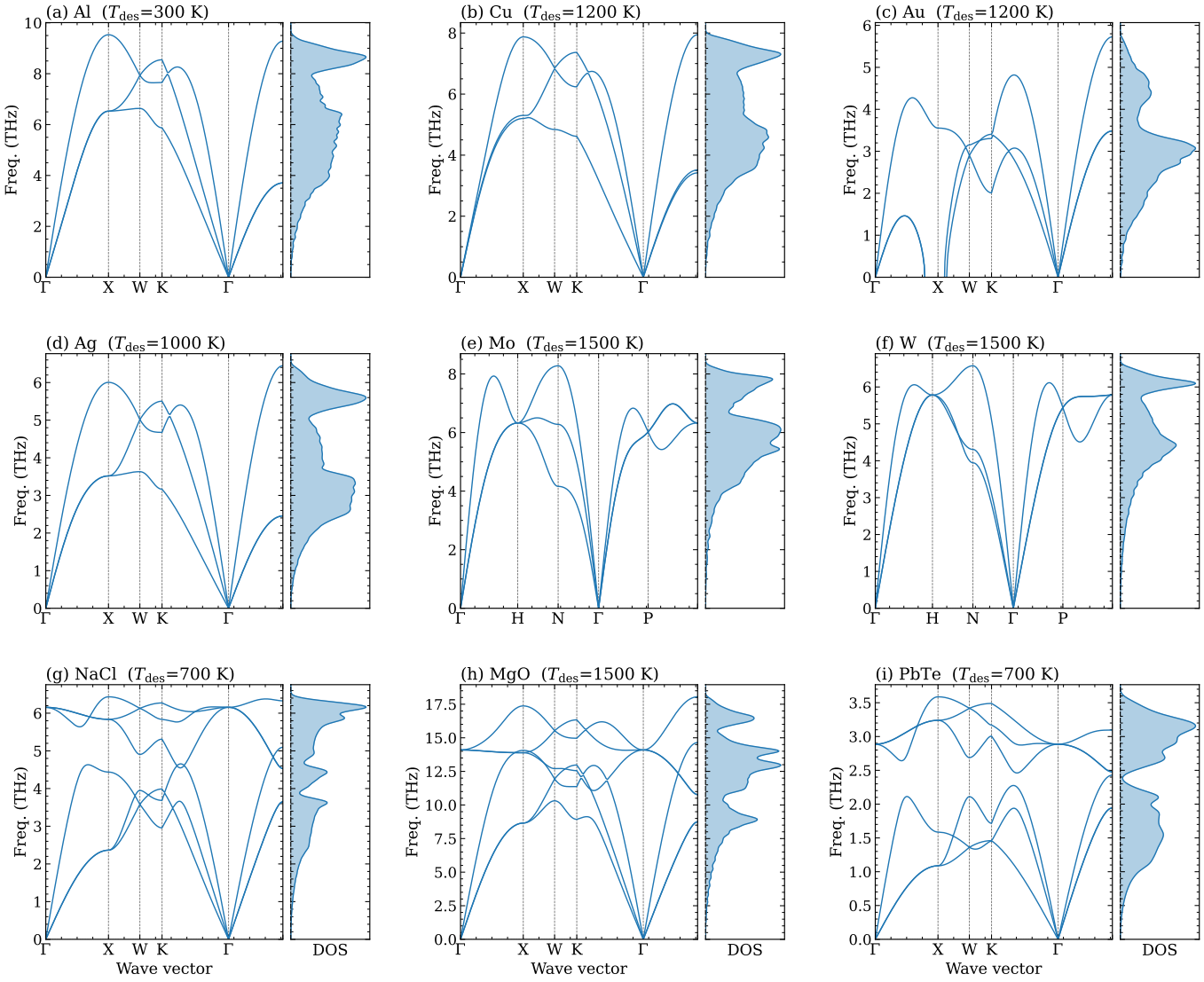


FIG. 1. SOTC effective-FC2 phonon dispersions and DOS for (a) Al (300 K), (b) Cu (1200 K), (c) Au (1200 K), (d) Ag (1000 K), (e) Mo (1500 K), (f) W (1500 K), (g) NaCl (700 K), (h) MgO (1500 K), and (i) PbTe (700 K). The stored band paths are Γ-X-W-K-Γ-L for FCC and rocksalt and Γ-H-N-Γ-P-H for BCC. DOS data use a $30 \times 30 \times 30$ q mesh and 0.1 THz broadening. Experimental neutron points are not overlaid in this archived figure. The three polar materials omit the non-analytic LO–TO correction, and the Au panel contains an instability; the panels are consequently interpreted qualitatively.

NIST–JANAF [20]. These fixed-volume results use one effective-FC2 model selected at T_{des}; they do not include thermal expansion, electronic heat capacity, or explicit higher-order anharmonic contributions. The comparison is restricted to SOTC and experiment because available external-method series lack matched settings and calculation provenance.

T_D^{SOTC} is the second-moment spectral value from the postprocessed IFC dataset, C_V^{SOTC} is evaluated at T_{des}, and ZPE is reported per formula-unit atom. Spectral, caloric, and ultrasonic Debye temperatures are not interchangeable; their differing definitions and measurement temperatures limit a point-by-point comparison.

The archived runner and postprocessed thermal summaries contain different T_D values for some materials. Table II records the postprocessed FC2 dataset, and its spectral Debye columns are interpreted descriptively. The heat-capacity and ZPE columns refer consistently to that postprocessed set.

At high temperature the harmonic phonon expression approaches $3R$, whereas the calorimetric column is C_p, not C_V, and can include expansion, electronic, and explicit anharmonic contributions. The tabulated aggregate values are therefore a descriptive benchmark; an integrated moment cannot determine the accuracy of individual branches. In particular, the favorable Au moment does not resolve the instability in its dispersion, and branch-resolved error statistics are unavailable for

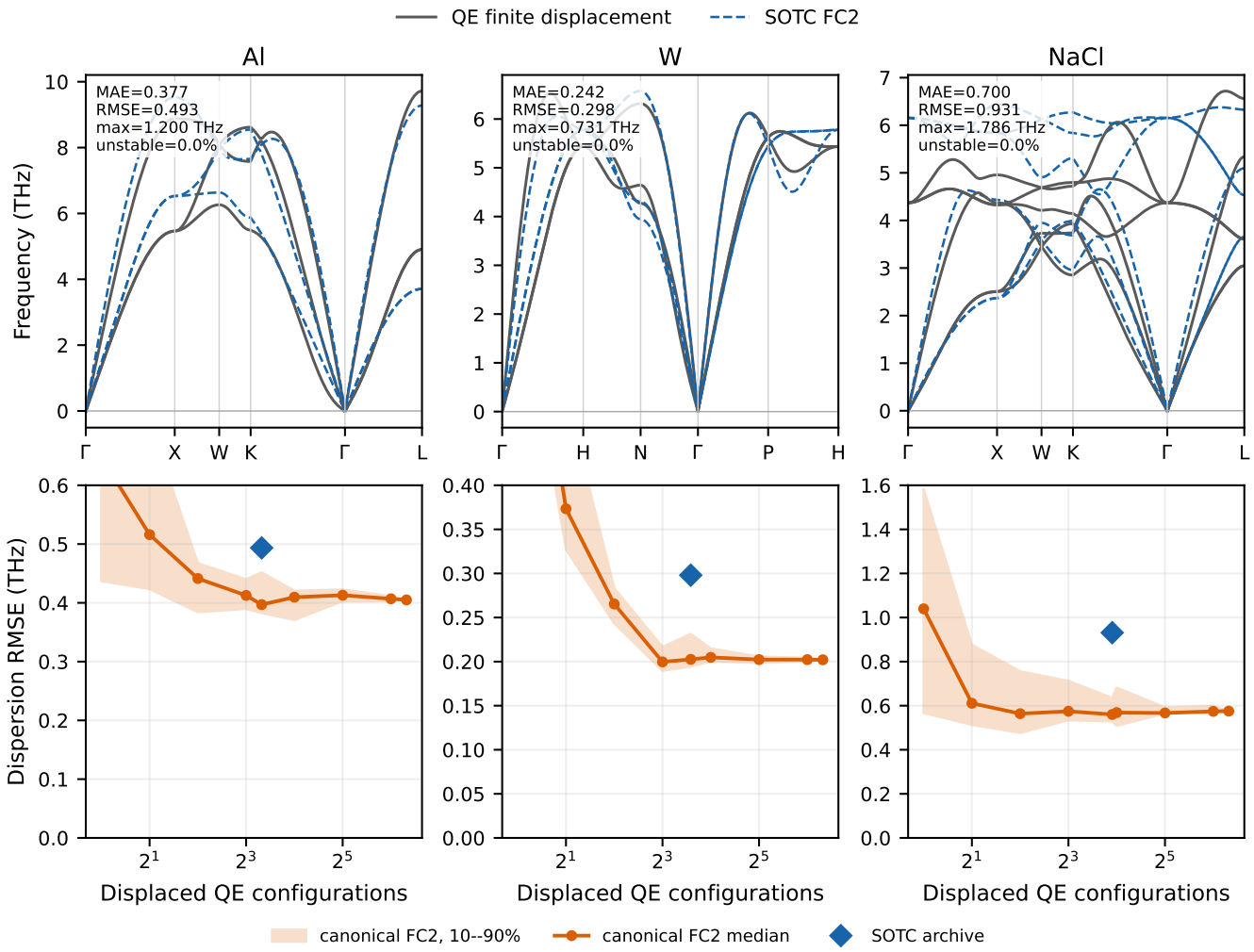


FIG. 2. Direct dispersion and sampling controls from archived QE calculations. Top: SOTC effective-FC2 bands (blue dashed) and independent conventional finite-displacement QE FC2 references (gray) for Al, W, and NaCl. Errors are pointwise sorted-mode MAE, RMSE, and maximum absolute error on the same reciprocal path. Bottom: median and 10–90% range over ten quantum-canonical subsets; the SOTC points use the same numbers of displaced configurations (10, 12, and 15). The canonical refits use the primary cutoff, ridge coefficient, central-force projection, and acoustic-sum-rule construction. The campaigns nevertheless use different cells, displacement ensembles, and electronic meshes, so this is a cross-archive control rather than a matched proof of sampling superiority or cost reduction.

the primary dataset.

D. Zero-Point Energies and Vibrational Free Energies

The zero-point energies (ZPE) and vibrational free energies in Table II are computed from the stored phonon DOS via Eqs. (10)–(11) (harmonic phonon formulas at the equilibrium volume V_0).

The zero-point energies (Table II) range from 0.020 eV/atom (Au, heavy mass and soft phonons) to 0.153 eV/atom (MgO, light mass and hard phonons), consistent with ZPE being the frequency-weighted half-quantum sum.

The vibrational entropy $S_{\mathrm{vib}}(T)$ at 1000 K spans 57–122 J/mol·K across the benchmark set, in the expected order (soft/heavy PbTe, Au > NaCl > Cu > Mo > Al > hard MgO). For aluminum, the fixed-volume phonon result is $S_{\mathrm{vib}}^{\mathrm{SOTC}}(1000\,\mathrm{K}) = 57$ J/(mol·K). A calorimetric entropy inferred from C_p/T is not a like-for-like validation because it also contains expansion, electronic, and explicit anharmonic terms. Because entropy is a DOS-weighted integral, it cannot by itself establish branch-resolved dispersion accuracy; the direct frequency controls provide the stricter comparison.

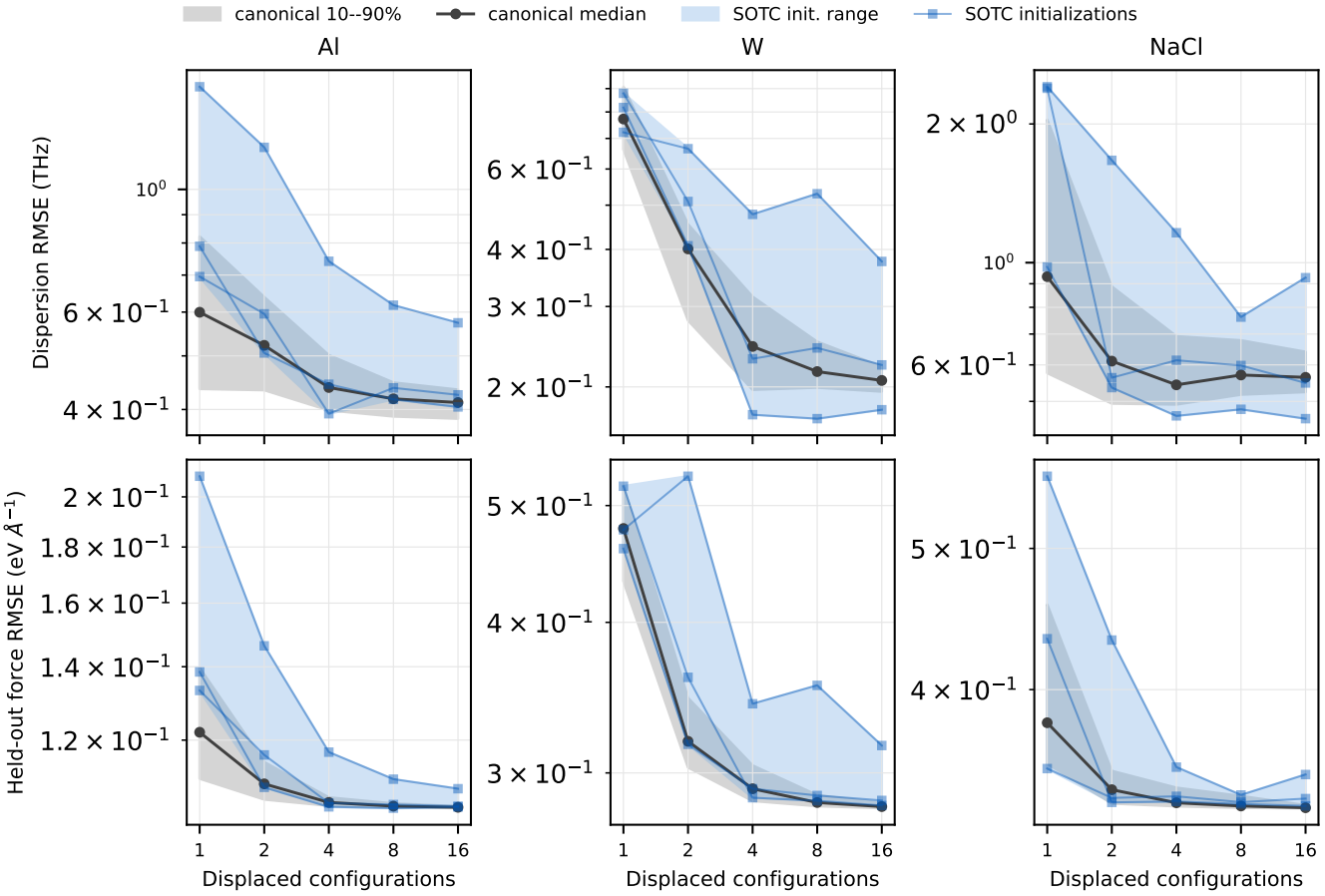


FIG. 3. Matched displacement-selection ablation for Al, W, and NaCl. Left: pointwise sorted-mode dispersion RMSE relative to the conventional finite-displacement FC2 reference. Right: force RMSE on the same 20 held-out canonical snapshots. Black curves and bands are the median and 10–90% range over 20 seeded canonical-Gaussian permutations; colored curves are three independently optimized correlator-matched sequences initialized at $0.75T_D$, T_D, and $1.25T_D$. Cells, electronic settings, force processing, fit priors, held-out data, q paths, and displaced-force counts are identical. The test isolates initial displacement selection rather than the complete self-consistent SOTC outer loop.

TABLE II. Benchmark summary: computed spectral quantities and a descriptive comparison of fixed-volume phonon C_V with calorimetric C_p.

Material	Struct.	T_{des}	$\omega_{\mathrm{max}}^{\mathrm{SOTC}}$ (THz)	T_D^{SOTC} (K)	T_D^{expt} (K)	C_V^{SOTC} (J/mol·K)	C_p^{calor} (J/mol·K)	$100(C_V - C_p)/C_p$ (%)	ZPE (eV/atom)
Al	FCC	300 K	10.3	450	390–428 [22]	22.9	24.4	−6.1	0.038
Ag	FCC	1000 K	6.8	297	∼225 [23]	24.9	27.1	−8.3	0.024
Au	FCC	1200 K	6.5	282	165† [24]	24.9	27.1	−8.1	0.020
Cu	FCC	1200 K	9.2	401	315–343 [25, 26]	24.8	26.4	−5.9	0.034
Mo	BCC	1500 K	8.7	380	380–470 [27]	24.9	28.7	−13.3	0.035
W	BCC	1500 K	7.3	318	310–400 [28]	24.9	26.9	−7.5	0.029
NaCl	Rocksalt	700 K	7.2	300	321 [29]	49.5	51.7	−4.4	0.056
MgO	Rocksalt	1500 K	20.3	726	743–800 [30, 31]	49.2	49.2	0.0	0.153
PbTe	Rocksalt	700 K	4.2	128	130–160 [32, 33]	47.3	48.3	−2.2	0.025

E. Mean-Square Displacements

Equations (3) and (12) define the same one-component isotropic displacement parameter U_{iso}, with $B = 8\pi^2 U_{\mathrm{iso}}$. Reinterpreting the archived total three-dimensional MSD fields at 300 K gives $U_{\mathrm{iso}} = 0.01088$, 0.00605, and 0.00516 Å^2 for Al, Au, and Cu, respectively. The corresponding B values are 0.859, 0.477, and 0.407 Å^2. The Al value lies within the cited 0.010–0.011 Å^2 interval [39]; Au is 53–64% below the cited 0.013–0.017 Å^2 range [40], and Cu is 36–48% below the cited 0.008–0.010 Å^2 range [41]. Thus correcting the factor of three resolves the Al discrepancy but does not validate Au or Cu. The Au value is not assigned physical significance because its FC2 dispersion is unstable; the Cu error remains a limitation of the fixed-volume effective-FC2 result.

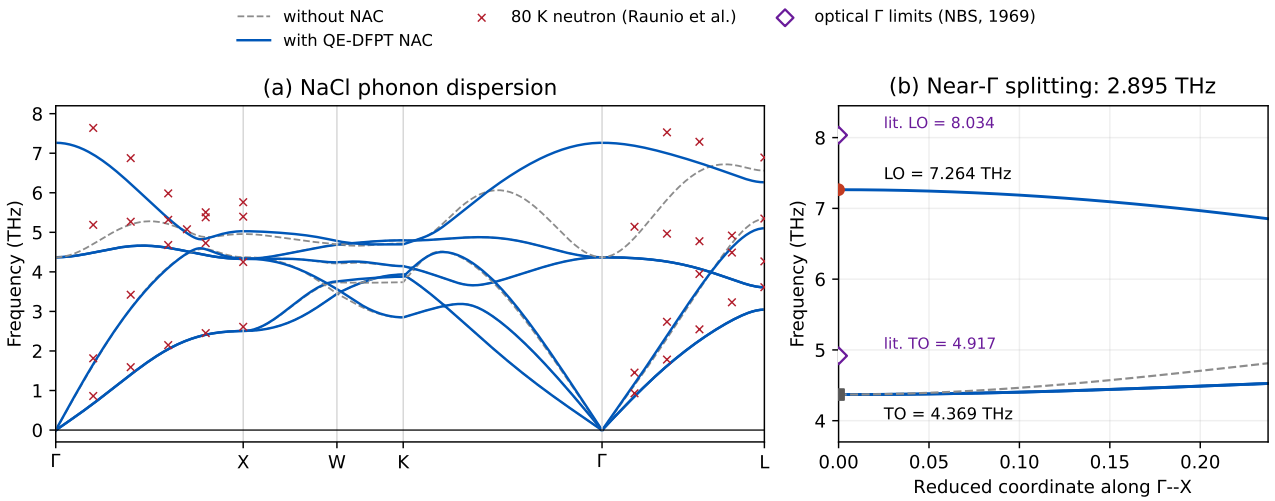


FIG. 4. NaCl LO–TO validation using completed QE-DFPT Born effective charges and electronic dielectric tensor. (a) Matched central-finite-displacement FC2 dispersion without NAC (gray dashed) and with NAC (blue). Red crosses are the 39 tabulated 80 K neutron points along Γ–X and Γ–L from Table I of Ref. [35], converted from angular frequency using $\nu = \omega/(2\pi)$. (b) Near-Γ optic branches along Γ–X. The calculated TO and LO limits are 4.369 and 7.264 THz, giving a 2.895 THz splitting; purple diamonds are the archival optical limits 4.917 and 8.034 THz [37]. The neutron data are temperature mismatched and incomplete along the full path, so no global experimental RMSE is assigned.

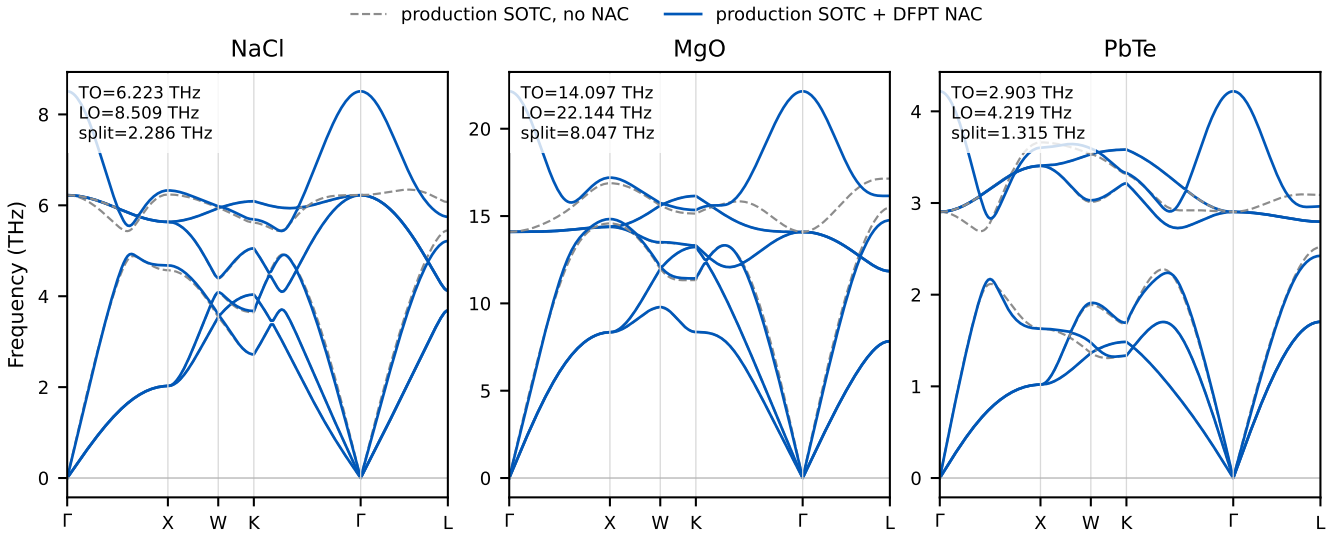


FIG. 5. Selected-production SOTC FC2 snapshot refits for NaCl, MgO, and PbTe without NAC (gray dashed) and with same-volume QE-DFPT NAC (blue). The directional Γ-point TO, LO, and splitting values are printed in each panel. The no-NAC Phonopy conversion reproduces the native refit frequencies to better than 2×10^{-6} THz RMSE.

F. Independent Higher-Order IFC Validation

We additionally tested whether the software can recover explicit higher-order force content in a workflow independent of the few-configuration SOTC benchmarks. Matched calculations were performed for FCC Al at 300 K, BCC W at 1500 K, and rocksalt NaCl at 700 K in $3 \times 3 \times 3$ primitive-cell supercells containing 27, 27, and 54 atoms, respectively. For each material, 50 configurations drawn from the quantum harmonic canonical distribution were paired with their exact inversions, giving 100 snapshots. Complete $+\mathbf{u}/-\mathbf{u}$ pairs were kept wholly within an 80-snapshot training set or a strictly held-out 20-snapshot test set. The independent QE settings were 40/320 Ry and a $4 \times 4 \times 4$ supercell k-mesh for Al, 30/240 Ry and $4 \times 4 \times 4$ for W, and 60/480 Ry and $2 \times 2 \times 2$ for NaCl. All used PBE, the pseudopotentials listed in Appendix J, a 10^{-9} Ry SCF threshold, 0.02 Ry

Marzari–Vanderbilt smearing for the metals, and fixed occupations for NaCl.

As a conventional harmonic baseline, central finite-displacement FC2 tensors were generated with phonopy from matched QE calculations in the same supercells, using +0.01 and –0.01 Å displacements and explicit subtraction of the undisplaced reference force. The thermal snapshots were then fit with the same symmetry-reduced design, successively retaining FC2, FC2+FC3, and FC2+FC3+FC4 terms. Space-group, index-permutation, translation, and acoustic-sum-rule constraints were imposed without ridge regularization. Table III evaluates every model on the unchanged test set. These are predictive force tests rather than comparisons of training residuals.

Temperature-fitting FC2 alone gives only modest improvements over the finite-displacement harmonic baseline: 3.3% for Al, 6.4% for NaCl, and 7.5% for W. Adding FC3 lowers the T-fit FC2 error by 72.7%, 43.0%, and 68.7%, respectively; adding FC4 then lowers the remaining error by 76.1%, 48.1%, and 56.8%. Relative to the conventional FD FC2 baseline, the final reductions are 93.7%, 72.3%, and 87.5%. The final held-out R^2 values are 0.99974, 0.96907, and 0.99943. All normalized designs have full column rank (condition numbers 312.7, 36.9, and 269.4), and the largest compact acoustic-sum-rule residual is 1.2×10^{-12} in the tensor's native units.

Snapshot-count fits using 20, 40, 60, and 80 training configurations, cutoff brackets of ±0.5 Å, and selected plane-wave-cutoff and k-mesh calculations were archived. The Al cutoff test selected 5.0 Å for FC2/FC3: relative to 4.5 Å, its held-out RMSE falls from 0.03103 to 0.00651 eV Å^{-1}. The denser-k-mesh force shift for Al remains larger than this final fit error. The Al tensors are therefore interpreted as force-regression results rather than electronically converged transport coefficients. NaCl has the largest sampled displacement amplitudes and the weakest held-out score; its result establishes substantial higher-order force content but not converged transport properties.

These fits quantify force regression rather than linewidths, lifetimes, lattice thermal conductivity, or higher-order free energies. The 100-snapshot canonical campaign validates higher-order IFC regression independently of the few-configuration SOTC design. File formats, generating scripts, and cross-format checks are documented in Appendix J.

V. DISCUSSION

A. Physical Validity of the Effective Harmonic Approximation

The primary SOTC self-consistency workflow fits a second-order effective-harmonic model. It can represent temperature-dependent curvature only insofar as the DFT force data and self-consistency procedure encode it in the fitted IFCs. The independent workflow in Sec. IV F fits explicit FC3 and FC4 tensors from a much larger canonical ensemble, but no linewidth, lifetime, transport coefficient, non-Gaussian nuclear distribution, or explicitly anharmonic free energy is computed here. Consequently, agreement of C_V, ZPE, or a Debye moment with experiment cannot by itself establish an accurate branch-resolved dispersion or an anharmonic variational solution: these are DOS-integrated observables that can hide compensating frequency errors.

The archived benchmarks show that the scalar correlator stopping metric becomes smaller than the chosen threshold in 2–5 iterations. That observation is limited to empirical convergence of the selected scalar metric. The finite fit also relies on the IFC cutoff, space-group and central-force constraints, the acoustic sum rule, ridge regularization, the initialization, and 1–8 configurations per iteration. Their separate contributions are not isolated by the primary archive. Figure 3 supplies a cell-, setting-, fit-, count-, and test-set-matched first-stage ablation for Al, W, and NaCl. It shows comparable nominal-initialization errors, improvements for the favorable $1.25T_D$ initialization, and deterioration for $0.75T_D$. This isolates initial displacement selection and directly measures Debye-initialization sensitivity, but it does not isolate every optimizer choice or establish convergence of the complete self-consistent outer loop. Complete SVD diagnostics remain unavailable for the primary runs.

The Au band audit finds a minimum of −1.066 THz at X [reduced q point $(0.5, 0, 0.5)$] on the lowest sorted branch. Modes below −0.05 THz occupy 45 of 753 plotted branch/q-point values (5.98%), spanning path indices 40–63 around X. The mesh instability fraction increases from 0.39% in iteration 1 to 15.6% in iteration 2, so self-consistency did not resolve it. Dependence on cutoff, cell, symmetry constraints, regularization, electronic convergence, and volume remains unseparated, and Au is excluded from the stable high-temperature dispersion assessment. The immutable polar arrays in Fig. 1 omit NAC; corrected selected-snapshot refits are shown in Fig. 5.

B. Measured Force-Call Cost

The primary SOTC runs used 2–32 displaced structures and one reference per iteration, for totals of 4–36 QE force calculations per material (Table I). These measured counts quantify workflow cost. Figure 3 compares the same dispersion and held-out-force observables with canonical-Gaussian sampling at equal displaced-force counts, using matched cells and electronic and fitting settings. At 16 calls, nominal-SOTC dispersion RMSE differs from the canonical median by (+3.2%), (+8.1%), and (-2.8%) for Al, W, and NaCl. Across all three pre-specified initializations, the relative changes span (-1.8%) to (+39.3%), (-13.7%) to (+82.2%), and

TABLE III. Matched model comparison for real-material higher-order IFC validation. RMSE is the Cartesian force-component error in eV Å^{-1} on 20 strictly held-out canonical snapshots. *FD FC2* is the conventional central finite-displacement phonopy model; *T-fit* models are fit to the 80 thermal training snapshots. p/r is the number of symmetry-reduced parameters and normalized design rank for the final FC2+FC3+FC4 fit. The final column is the RMSE reduction relative to FD FC2.

Material	T (K)	atoms	p/r	FD FC2	T-fit FC2	T-fit FC2+FC3	T-fit FC2+FC3+FC4	reduction (%)
Al	300	27	106/106	0.10294	0.09955	0.02721	0.00651	93.7
NaCl	700	54	48/48	0.35432	0.33152	0.18906	0.09805	72.3
W	1500	27	126/126	0.29816	0.27566	0.08634	0.03730	87.5

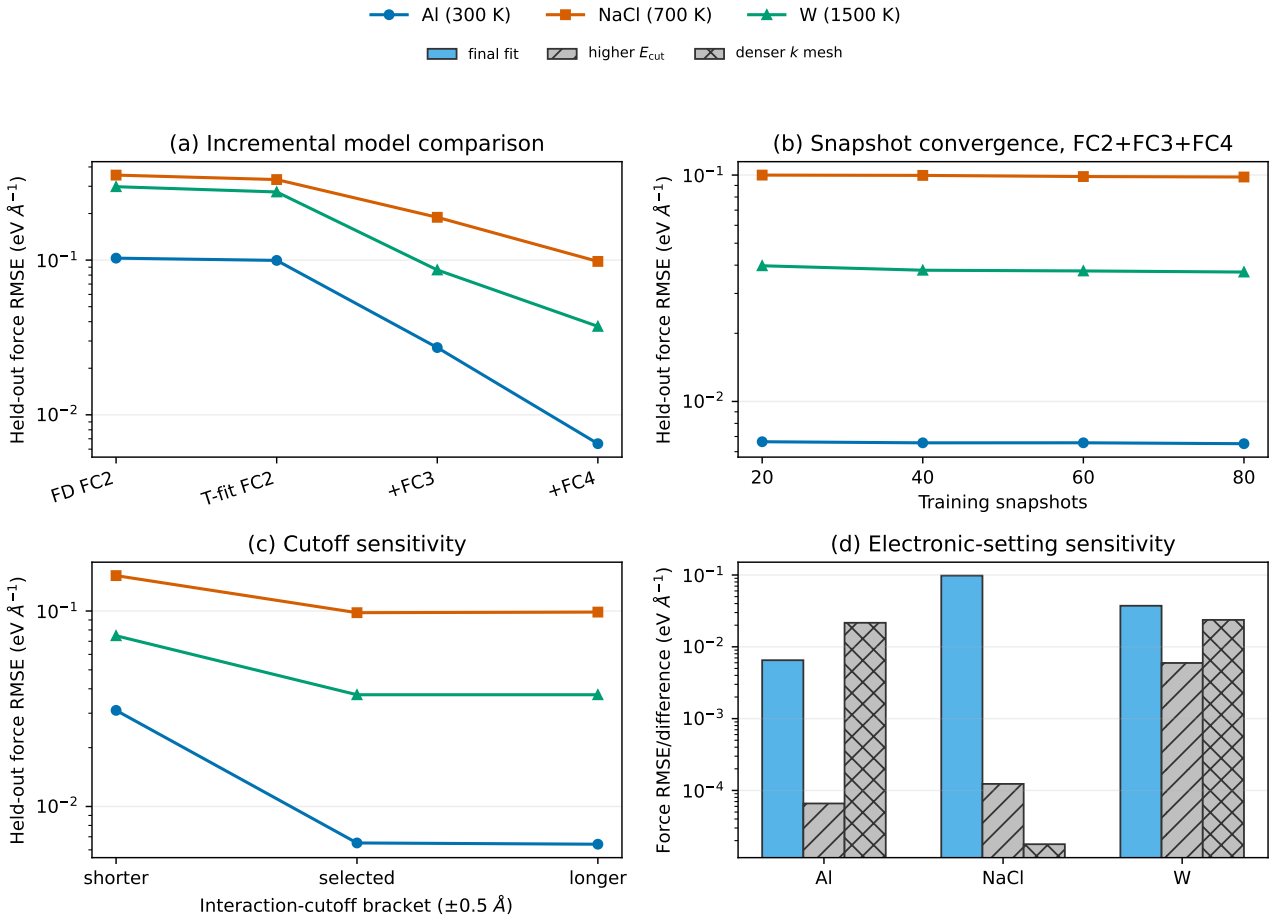


FIG. 6. Post-IFC validation for the completed QE force campaign. (a) Held-out force-component RMSE for conventional finite-displacement FC2, temperature-fitted FC2, FC2+FC3, and FC2+FC3+FC4 models. (b) Final-model convergence with the number of thermal training snapshots. (c) Sensitivity to interaction cutoffs bracketed by ±0.5 Å around the selected values. (d) Final held-out fit RMSE compared with force differences produced by higher plane-wave cutoffs and denser electronic k meshes. Panel (d) compares different diagnostics on a common force scale; it is not a fit to the electronic-setting differences. All predictive errors use the unchanged 20-snapshot test set.

(-18.7%) to (+64.5%), respectively, where negative values are improvements. Because the outcome depends strongly on initialization and wall times were not normalized, these results are reported as an error-versus-force-call comparison rather than a universal sampling-efficiency factor or computational speedup.

The independent higher-order campaign provides one controlled cost comparison. The conventional central finite-displacement FC2 references required three QE calculations for Al and W (one reference plus two displaced states) and five for NaCl (one plus four). The complete validation campaigns used 100 additional canonical-snapshot calculations and six electronic-convergence calculations, for totals of 109, 111, and 109 calls. Thus the completed campaigns cost 36.3, 22.2, and 36.3 times the corresponding finite-displacement FC2 call counts. That additional cost supplies training and held-out evidence for FC3/FC4 and reduces force RMSE by 72–

94%. Because this campaign uses canonical sampling, its cost is separate from the SOTC configuration-design cost. Once the forces are available, fitting FC2, FC2+FC3, and FC2+FC3+FC4 uses the same 80 training snapshots and requires no additional DFT calculations.

Force-call counts serve as the cost metric because archived wall times use different processor allocations and omit post-processing on a common basis. The comparison therefore consists of the absolute counts and the matched campaign ratios above.

C. Comparison with Related Methods

The temperature-dependent effective potential (TDEP) framework fits effective second- and higher-order IFCs to finite-temperature force data; molecular dynamics, Gaussian sampling, and other ensembles can supply those data [5–8]. The SCHA is a variational effective-harmonic theory, and SSCHA performs its stochastic sampling and minimization [9–11]. Under consistent ensembles and converged estimators, stochastic TDEP and SSCHA can reduce to the same effective-harmonic stationary solution and free energy; therefore absence of variational minimization is not used here as a blanket physical distinction. Mode-coupling approaches provide a distinct non-perturbative route to renormalized lattice dynamics [12], while AIMD generates classical nuclear trajectories on a first-principles potential-energy surface [13].

In IFC-rank terms, the primary SOTC workflow and every dispersion in Fig. 1 use an effective FC2 model; the conventional phonopy and LO–TO control in Fig. 4 also use FC2. The independent campaign in Fig. 6 explicitly compares FC2, FC2+FC3, and FC2+FC3+FC4 fits. The cited TDEP literature includes effective FC2 treatments [5, 6] and an explicit effective-FC3 formulation [8]. SSCHA's optimized trial Hamiltonian is harmonic (FC2), although its free-energy derivatives and spectra contain anharmonic effects; it should not be described as an explicit FC3/FC4 tensor fit. Mode-coupling theory is likewise not classified by a single fitted IFC rank.

SOTC has a narrower objective: it optimizes a finite set of spatial displacement moments before evaluating forces and then fits effective FC2. Its sampling advantage must be assessed with matched controls at the same physical level, cell, temperature, electronic settings, and target observable. The present benchmarks do not compare explicit anharmonic free energies or renormalized spectra with the established methods above. The special displacement method [42] is a particularly relevant deterministic comparison.

D. Limitations and Validation Boundaries

Sampling and identifiability. The scalar stopping metric tests selected shell-averaged displacement correlators; it does not test every tensor component or guarantee convergence of the full phonon dispersion. With 1–8 displaced configurations per iteration and ridge regularization, scalar convergence and unique FC2 recovery are distinct properties. The tested domain is restricted to the reported cells, temperatures, cutoffs, and electronic settings. The matched controls in Fig. 3 compare pointwise sorted-mode frequencies and held-out forces at the same displaced-configuration counts, cells, electronic settings, and FC2 fitting choices. They show that the best tested initialization can beat the canonical median, but the low-T_D initialization can be substantially worse. Only three coupled initialization/optimizer starts and the initial design stage were tested; a universal or initialization-independent sampling advantage is therefore not established.

Long-range electrostatics. Polar insulators require a non-analytic correction to the dynamical matrix [43]:

$$D_{ij}^{\alpha\beta,\mathrm{NAC}}(\hat{\mathbf{q}}) = \frac{4\pi e^2}{\Omega_0\sqrt{M_i M_j}} \frac{(\sum_\gamma Z_i^{*\alpha\gamma}\hat{q}_\gamma)(\sum_\delta Z_j^{*\beta\delta}\hat{q}_\delta)}{\sum_{\gamma\delta}\hat{q}_\gamma\varepsilon_\infty^{\gamma\delta}\hat{q}_\delta}.$$

Here Ω_0, Z^*, and ε_∞ are the primitive-cell volume, Born effective-charge tensor, and clamped-ion dielectric tensor. Figure 4 applies QE DFPT tensors in an independent matched NaCl control, while Fig. 5 applies same-volume tensors to selected-production snapshot refits for all three polar compounds. The latter refits are not asserted to reproduce the immutable no-NAC arrays because the selected in-memory FC2 objects were not serialized. Full experimental branch-resolved validation remains limited to NaCl, and the scalar-relativistic PbTe result does not establish spin–orbit insensitivity.

Higher-order and transport scope. The independent canonical campaign in Sec. IV F validates constrained FC3 and FC4 fitting using 100 snapshots. It does not show that the few-configuration SOTC design identifies FC3 or FC4, and it does not report phonon linewidths, lifetimes, or lattice thermal conductivity. A transport extension requires convergence of reciprocal-space integration, scattering phase space, isotope and boundary models, higher-order cutoffs, and training-set size [44].

Structural-instability scope. The current harmonic trial correlator requires a stable reference spectrum and cannot represent a non-Gaussian or double-well displacement distribution. Systems near displacive phase transitions are therefore outside the demonstrated scope. Extending the method to this regime would require an explicitly anharmonic reference distribution and validation of both free energies and renormalized spectra against established ensemble methods.

Thermodynamic and magnetic scope. The reported thermodynamic quantities are fixed-volume effective-FC2 values of C_V, ZPE, F_{vib}, and S_{vib}. Thermal expansion and isobaric heat capacity are not evaluated; they require direct static energies and phonons over a

converged volume grid. Magnetic materials also require spin–lattice coupling, which is absent from the present model [45].

Unresolved validation boundaries. The matched ablation covers Al, W, and NaCl at one temperature and volume each and tests three coupled Debye-target/optimizer initializations only for first-stage displacement selection; it does not establish full outer-loop or broad chemical transferability. Complete primary-run SVD diagnostics and a resolved Au stability test remain unavailable. The corrected MSD benchmark validates Al but not Au or Cu. These boundaries limit the supported conclusions to the explicit effective-FC2 and independent higher-order tests reported above.

VI. CONCLUSIONS

SOTC selects collective displacements by matching prescribed spatial correlators and uses the resulting forces in a constrained effective-FC2 fit. Across the nine primary benchmarks, the scalar stopping criterion is reached in 2–5 iterations and 4–36 force calculations including references. These counts describe workflow cost; FC2 uniqueness, dispersion convergence, and sampling efficiency are separate properties.

The fixed-volume FC2 heat capacities approach the harmonic Dulong–Petit limit at high temperature. Their tabulated comparison with calorimetric C_p is descriptive rather than a direct accuracy test. The matched displacement-selection ablation gives 16-call canonical median dispersion RMSEs of 0.412, 0.207, and 0.564 THz for Al, W, and NaCl. The pre-specified nominal-T_D designs give 0.425, 0.223, and 0.548 THz; nominal force errors are likewise essentially the canonical medians. Across all three initializations, dispersion RMSE spans 0.404–0.574, 0.178–0.377, and 0.459–0.928 THz. Thus nominal performance is comparable and a favorable initialization can improve the tested metric, but a universal or initialization-independent advantage and a wall-time speedup are not established. A separate 100-snapshot canonical campaign shows that constrained FC2+FC3+FC4 fits lower held-out force RMSE by 72–94% relative to matched finite-displacement FC2 baselines for Al, NaCl, and W. It validates higher-order regression with ensemble forces, not recovery of FC3/FC4 from the few-configuration SOTC workflow.

Same-volume DFPT tensors applied to selected-production FC2 refits give LO–TO splittings of 2.286, 8.047, and 1.315 THz for NaCl, MgO, and PbTe. The independent conventional-FC2 NaCl control gives 2.895 THz and retains the optical and neutron comparison. Full experimental branch-resolved validation is not claimed for MgO or scalar-relativistic PbTe. The Au instability prevents its use as evidence of a stable dispersion, and the corrected 300 K displacement parameter agrees with the cited interval for Al but is 53–64% low for Au and 36–48% low for Cu. The factor-of-three normalization error is resolved, but those Au and Cu discrepancies remain explicit limitations.

Within these limits, SOTC is applicable to reproducible effective-FC2 model construction for stable, nonmagnetic crystals and to fixed-volume phonon-DOS thermodynamics such as C_V, ZPE, F_{vib}, and S_{vib}. It can also provide a deterministic harmonic reference for larger finite-temperature workflows. The independently tested regression route is applicable to FC3/FC4 extraction when a sufficiently large ensemble of forces is available. Strongly non-Gaussian nuclear motion, anharmonic free energies, phonon lifetimes, thermal conductivity, and computational acceleration lie outside the demonstrated application domain.

ACKNOWLEDGMENTS

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Appendix A: Derivation of the Quantum Thermal Displacement Correlator

We derive Eq. (1) from the quantum harmonic lattice. In the second-quantized description, the displacement of atom i of mass M_i at lattice site $\mathbf{R}$ along Cartesian direction α is:

$$\hat{u}_i^\alpha(\mathbf{R}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{q},s} \sqrt{\frac{\hbar}{2M_i\omega_{\mathbf{q}s}}} e_i^\alpha(\mathbf{q}s)\big(\hat{a}_{\mathbf{q}s} + \hat{a}^\dagger_{-\mathbf{q}s}\big) e^{i\mathbf{q}\cdot\mathbf{R}}$$

where $\hat{a}^\dagger_{\mathbf{q}s}$ ($\hat{a}_{\mathbf{q}s}$) creates (annihilates) a phonon in branch s at wavevector $\mathbf{q}$, $e_i^\alpha(\mathbf{q}s)$ is the polarization vector, and N is the number of unit cells.

Computing the thermal average. The thermal correlator between atoms i (at the origin) and j (at $\mathbf{R}$) is $C_{ij}^{\alpha\beta}(\mathbf{R},T) = \langle \hat{u}_i^\alpha(\mathbf{0})\hat{u}_j^\beta(\mathbf{R})\rangle_T$ with $\langle\cdot\rangle_T = \mathrm{Tr}(e^{-\hat{H}/k_BT}\cdot)/Z$. Inserting the mode expansion:

$$C_{ab}^{\alpha\beta}(\mathbf{R},T) = \frac{\hbar}{2N_c\sqrt{M_aM_b}} \sum_{\mathbf{q}s}\sum_{\mathbf{q}'s'} \frac{e_a^\alpha(\mathbf{q}s)e_b^\beta(\mathbf{q}'s')}{\sqrt{\omega_{\mathbf{q}s}\omega_{\mathbf{q}'s'}}} \times \mathcal{A}(\mathbf{q}s,\mathbf{q}'s')e^{i\mathbf{q}'\cdot\mathbf{R}}.$$

where the bosonic correlator $\mathcal{A}$ collects the four terms from $(\hat{a}+\hat{a}^\dagger)(\hat{a}'+\hat{a}'^\dagger)$. Using the standard equilibrium averages:

$$\langle \hat{a}^\dagger_{\mathbf{q}s}\hat{a}_{\mathbf{q}'s'}\rangle_T = n_B(\omega_{\mathbf{q}s})\,\delta_{\mathbf{q}\mathbf{q}'}\delta_{ss'},$$
$$\langle \hat{a}_{\mathbf{q}s}\hat{a}^\dagger_{\mathbf{q}'s'}\rangle_T = \big(n_B(\omega_{\mathbf{q}s})+1\big)\delta_{\mathbf{q}\mathbf{q}'}\delta_{ss'}$$

where $n_B(\omega) = (e^{\hbar\omega/k_BT} - 1)^{-1}$ is the Bose-Einstein occupation, and noting that anomalous averages $\langle \hat{a}\hat{a}\rangle_T = \langle \hat{a}^\dagger \hat{a}^\dagger\rangle_T = 0$ vanish, we obtain:

$$\mathcal{A} = (2n_B + 1)\,\delta_{\mathbf{q},-\mathbf{q}'}\delta_{ss'}.$$

The coth **identity.** We prove $2n_B(\omega) + 1 = \coth(\hbar\omega/2k_BT)$. Starting from the definition, with $x = \hbar\omega/k_BT$:

$$2n_B + 1 = \frac{2}{e^x - 1} + 1 = \frac{2 + e^x - 1}{e^x - 1} = \frac{e^x + 1}{e^x - 1}$$

Multiply numerator and denominator by $e^{-x/2}$:

$$\frac{e^{x/2} + e^{-x/2}}{e^{x/2} - e^{-x/2}} = \frac{\cosh(x/2)}{\sinh(x/2)} = \coth\left(\frac{\hbar\omega}{2k_BT}\right). \qquad \square$$

Using $\delta_{\mathbf{q},-\mathbf{q}'}$ to collapse the $\mathbf{q}'$ sum, using $e_i^{\alpha *}(\mathbf{q}s) = e_i^\alpha(-\mathbf{q}s)$ (time-reversal symmetry), and substituting, we arrive at the final result:

$$C_{ab}^{\alpha\beta}(\mathbf{R}, T) = \frac{\hbar}{2N_c\sqrt{M_aM_b}} \sum_{\mathbf{q},s} \frac{e_a^{\alpha *}(\mathbf{q}s) e_b^\beta(\mathbf{q}s)}{\omega_{\mathbf{q}s}} \times \coth\left(\frac{\hbar\omega_{\mathbf{q}s}}{2k_BT}\right) e^{-i\mathbf{q}\cdot\mathbf{R}}.$$

This is Eq. (1) with the same negative phase convention. At high temperature, $\coth(x/2) \to 2/x$ for $x = \hbar\omega/k_BT$, giving the classical mode-amplitude scaling k_BT/ω^2. At zero temperature, $\coth(x/2) \to 1$, leaving the zero-point contribution of every mode.

Appendix B: Distance Weighting and Correlation Range

The equal-time displacement correlator in Eq. (1) is a static canonical quantity. Its spatial behavior follows from the full low-frequency dispersion, eigenvectors, dimensionality, and boundary conditions. A finite dynamical phonon lifetime broadens a spectral function, but does not by itself imply a universal exponential bound on this equal-time correlator. In particular, acoustic modes can produce long-ranged or algebraic contributions. The relations $\tau = \hbar/(\gamma k_BT)$, $\xi_T = \bar{v}_s\tau$, and $K_{\min} \propto T^{-3}$ omit the anharmonic matrix elements and phase space entering a three-phonon rate; no such scaling is assumed.

The exponential $w(R;\xi_w)$ in Eq. (5) is retained only as a numerical weighting choice. It is not interpreted as a phonon mean free path or a proven thermal coherence length. The real-space IFC cutoff, supercell dimensions, and weighting length must instead be converged directly against branch-resolved frequencies. The archive contains no material-specific comparison of $C_2(R,T)$ with exponential and non-exponential decay models.

Appendix C: The SOTC Objective Function: Definition and Gradient

For the finite vector of supercell displacements, $\mathcal{Q}$ is a multivariable objective function rather than a functional. In the species-resolved shell notation of Eq. (2), it is

$$\mathcal{Q}(\mathbf{u}) = \sum_{a,b} \sum_{R<r_{\text{cut}}} w_{ab}(R;\xi_w)\Big[\bar{C}_{2,ab}^{\text{cell}}(R;\mathbf{u}) - \bar{C}_{2,ab}(R,T)\Big]^2,$$

$$w_{ab}(R;\xi_w) = \exp(-R/\xi_w).$$

For the numerical objective, ideal shell membership is approximated by rounding pair distances to 0.05 Å bins while retaining ordered basis/species-pair channels. The bin width is an explicit input and is distinct from a space-group-orbit construction. The weight is a tunable numerical emphasis, not a formally optimal envelope derived from a phonon lifetime. Other positive weights are admissible, and ξ_w must be included in cutoff/sensitivity tests. The scalar shell objective also omits off-diagonal and anisotropic tensor components; extending the target to the full $C_{ab}^{\alpha\beta}(\mathbf{R},T)$ is possible but requires a correspondingly larger identifiability analysis.

Expansion in the phonon basis. The displacement field on the supercell of n atoms is expanded in phonon eigenvectors:

$$u_i^\alpha = \frac{1}{\sqrt{M_i}} \sum_{\mathbf{q}s} \left[Q_{\mathbf{q}s}\, e_i^\alpha(\mathbf{q}s)\, e^{i\mathbf{q}\cdot\mathbf{R}_i^0} + \text{c.c.}\right]$$

where $Q_{\mathbf{q}s}$ are complex mode amplitudes. The SOTC instantaneous correlator then reads:

$$\hat{C}_{ij}^{\alpha\beta}(\mathbf{R}) = \frac{1}{\sqrt{M_iM_j}} \sum_{\mathbf{q}s,\mathbf{q}'s'} Q_{\mathbf{q}s}Q_{\mathbf{q}'s'}^*\, e_i^\alpha(\mathbf{q}s)\, e_j^{\beta *}(\mathbf{q}'s')\, e^{i(\mathbf{q}-\mathbf{q}')\cdot\mathbf{R}_j^0}$$

Gradient of $\mathcal{Q}$. Differentiating with respect to the real mode amplitude $A_{\mathbf{q}s} = |Q_{\mathbf{q}s}|$:

$$\frac{\partial\mathcal{Q}}{\partial A_{\mathbf{q}s}} = 4 \sum_{(i,j),\mathbf{R}} w(|\mathbf{R}|)\big[\hat{C}_{ij}^{\alpha\beta}(\mathbf{R}) - C_{ij}^{\alpha\beta}(\mathbf{R},T)\big] \frac{\partial\hat{C}_{ij}^{\alpha\beta}}{\partial A_{\mathbf{q}s}}$$

Initial amplitude. The harmonic mode amplitude provides an initialization:

$$A_{\mathbf{q}s}^{harm}(T) = \sqrt{\frac{\hbar}{2\omega_{\mathbf{q}s}} \coth\left(\frac{\hbar\omega_{\mathbf{q}s}}{2k_BT}\right)}$$

Matching individual harmonic mode variances does not make one deterministic configuration reproduce the complete thermal covariance, nor does it prove that this point is stationary for the finite shell objective. Optimization is therefore required even for a commensurate mesh.

The relevant numerical diagnostics are the final objective value and its sensitivity to initialization and optimizer settings; gradient descent from $A_{\mathbf{q}s}^{harm}$ is an algorithmic choice rather than an exactness result.

Appendix D: IFC Extraction by Ridge Regression: Full Derivation

Building the regression matrix. For each atom pair (i, j) with separation $\mathbf{R}$ within r_{cut}, Newton's third law gives $\mathbf{F}_i = -\sum_j \mathbf{\Phi}_{ij}(\mathbf{u}_j - \mathbf{u}_i)$ for collective SOTC configurations (where all atoms are simultaneously displaced). This relative-displacement form is algebraically equivalent to $-\sum_j \mathbf{\Phi}_{ij}\mathbf{u}_j$ when the acoustic sum rule $\sum_j \mathbf{\Phi}_{ij} = \mathbf{0}$ holds exactly, but is numerically more stable for collective displacements because it prevents spurious rigid-translation forces from contaminating the regression. Vectorizing over all n atoms and 3 directions yields:

$$\Delta\mathbf{F} = -\mathbf{G}\,\mathbf{p}.$$

Here $\Delta\mathbf{F} \in \mathbb{R}^{3nN_{\text{ens}}}$ stacks the reference-subtracted forces from all configurations, $\mathbf{p} \in \mathbb{R}^{N_{\text{IFC}}}$ contains the independent symmetry-reduced parameters, and $\mathbf{G} \in \mathbb{R}^{3nN_{\text{ens}} \times N_{\text{IFC}}}$. Before symmetry aggregation, a contribution associated with $p = \Phi_{ij}^{\alpha\beta}$ is proportional to the relative displacement $u_j^\beta - u_i^\beta$, not u_j^β alone.

Normal equations and ridge regularization. The least-squares solution minimizes $\|\mathbf{G}\mathbf{p} + \Delta\mathbf{F}\|^2$. When $N_{\text{IFC}} > 3n$ (underdetermined) or $\mathbf{G}^T\mathbf{G}$ is near-singular (correlated IFC parameters), Tikhonov regularization adds a penalty $\lambda\|\mathbf{p}\|^2$:

$$\begin{aligned}\mathbf{p}^* &= \arg\min_{\mathbf{p}} \left(\|\mathbf{G}\mathbf{p} + \Delta\mathbf{F}\|^2 + \lambda\|\mathbf{p}\|^2\right) \\ &= -\left(\mathbf{G}^T\mathbf{G} + \lambda\mathbf{I}\right)^{-1}\mathbf{G}^T\Delta\mathbf{F}\end{aligned}$$

Rank and condition diagnostics. The singular values of the IFC design matrix are not, in general, identical to the normal-mode amplitudes: the matrix also contains bond geometry, parameter sharing, symmetry projection, and the stacking of configurations. No analytical condition-number advantage over a random/Gaussian control follows from exciting every phonon mode. Rank, singular values, and condition number must be computed from each actual symmetry-reduced matrix with a stated threshold. The effect of ridge regularization is reported through sensitivity to λ and error against a converged reference; it is not assumed to be unbiased.

Acoustic sum rule enforcement. The ASR, $\sum_j \Phi_{ij}^{\alpha\beta} = 0$ for each i, α, β, is imposed as a hard linear constraint via the Lagrange-multiplier method. Write the constrained optimization:

$$\min_{\mathbf{p}} \|\mathbf{G}\mathbf{p} + \Delta\mathbf{F}\|^2 + \lambda\|\mathbf{p}\|^2 \quad \text{subject to} \quad \mathbf{A}_{\text{ASR}}\mathbf{p} = \mathbf{0}$$

Solution via the KKT conditions gives the projected system: let $\mathbf{N}$ be the null-space basis of $\mathbf{A}_{\text{ASR}}$ (computed once by QR decomposition), write $\mathbf{p} = \mathbf{N}\tilde{\mathbf{p}}$, and solve the unconstrained problem for $\tilde{\mathbf{p}}$ using the reduced matrix $\tilde{\mathbf{G}} = \mathbf{G}\mathbf{N}$.

Central-force symmetry projection. For high-symmetry bonds (FCC nearest neighbors, rocksalt nearest neighbors), the IFC tensor can be decomposed into longitudinal and transverse components:

$$\Phi_{ij}^{\alpha\beta}(\hat{\mathbf{R}}) = A_{ij}\hat{R}_\alpha\hat{R}_\beta + B_{ij}\left(\delta_{\alpha\beta} - \hat{R}_\alpha\hat{R}_\beta\right)$$

where $\hat{\mathbf{R}} = \mathbf{R}/|\mathbf{R}|$ is the bond unit vector, A_{ij} is the longitudinal (bond-stretching) force constant, and B_{ij} is the transverse (bond-bending) force constant. This projection reduces the number of free IFC parameters per bond from 9 (general 3×3 tensor) to 2, improving the regression rank by a factor of $\sim$4.5 and dramatically stabilizing the fit.

Appendix E: Finite-Cell Truncation and IFC Identifiability

A finite SOTC cell limits accuracy through real-space IFC truncation, periodic-image aliasing, and identifiability of the force-constant fit. The first two errors are material dependent; polar crystals additionally require the non-analytic term in Sec. V.D. Because systematic cell- and cutoff-series are not available for every primary dispersion, those results are conditional on the reported choices.

For an n-atom cell and N_{ens} configurations, the force design has at most $3nN_{\text{ens}}$ independent rows. Symmetry and sum-rule constraints reduce the parameter count. Ridge regularization can stabilize a deficient fit but cannot increase its rank, and excitation of every normal mode does not by itself guarantee full column rank.

Table III reports reduced parameter counts, normalized rank, and condition number for the independent 100-snapshot campaign; all three final designs are full rank under their stated constraints. Equivalent saved matrices and SVD diagnostics are unavailable for the nine primary runs. Their scalar self-consistency measures numerical iteration convergence rather than unique FC2 recovery or representability.

Appendix F: Self-Consistency Map and Empirical Convergence

One outer iteration defines a numerical map $\mathcal{F} : C_2^{(n)} \mapsto C_2^{(n+1)}$: design configurations, evaluate forces,

fit constrained IFCs, construct phonons, and update the target correlators. Linear mixing forms

$$C_{2,\mathrm{mix}}^{(n+1)} = \beta\,\mathcal{F}(C_2^{(n)}) + (1-\beta)C_2^{(n)}.$$

The archived calculations stop when the norm of the change falls below 3×10^{-3} Å^2, subject to a material-specific minimum iteration count.

No contraction bound for $\mathcal{F}$ has been established. In particular, an estimate in terms of an effective Grüneisen parameter did not derive the derivatives of the displacement optimizer and regularized IFC fit, and cannot guarantee convergence. Numerical safeguards such as mixing, correlator floors, omission/rescaling of imaginary modes, and selection of a previous best iteration also modify the practical map. Accordingly, convergence of the scalar stopping metric must be distinguished from convergence to a unique physical IFC solution. Initialization, mixing, safeguard, and configuration-count sensitivity tests are required and are reported as empirical diagnostics rather than a fixed-point theorem.

Appendix G: Variational Scope of Deterministic Moment Matching

For a trial harmonic Hamiltonian H_0, the Gibbs–Bogoliubov inequality

$$F_{\mathrm{exact}} \le F_0 + \langle H - H_0\rangle_0$$

underlies variational self-consistent harmonic methods [9–11]. A method inherits this upper-bound property only if it evaluates the required expectation consistently and performs the appropriate variational minimization.

Matching a finite set of second moments,

$$\frac{1}{K}\sum_{k=1}^{K} u_i^{\alpha,(k)} u_j^{\beta,(k)} \simeq C_{ij}^{\alpha\beta},$$

does not by itself define a multidimensional Gauss–Hermite rule for general anharmonic forces or energies. It neither integrates all terms entering the Gibbs–Bogoliubov functional nor guarantees exponential quadrature convergence. The present SOTC algorithm also fits forces by ridge regression and stops on a correlator change; it does not minimize $F_0 + \langle H - H_0\rangle_0$. SOTC is therefore not assigned a rigorous variational upper bound, equivalence to a deterministic stochastic quadrature, or exponential convergence with configuration count.

The useful connection is limited to the use of harmonic covariance as organizing information for an effective harmonic model. Variational methods optimize a distribution, whereas SOTC selects finite configurations to match chosen moments and fits IFCs. Whether this deterministic moment design improves a force fit is an empirical question addressed by matched controls, not a consequence of the Gibbs–Bogoliubov inequality.

Appendix H: Harmonic Free Energy on the SOTC Cell

After self-consistency convergence, the renormalized frequencies $\omega^*_{\mathbf{q}s}(T)$ define the free energy of the best harmonic approximation. The single-mode partition function is:

$$\begin{aligned} Z_\omega &= \sum_{n=0}^{\infty} e^{-\hbar\omega(n+1/2)/k_BT} \\ &= e^{-\hbar\omega/2k_BT}\cdot\frac{1}{1-e^{-\hbar\omega/k_BT}} \\ &= \frac{1}{2\sinh(\hbar\omega/2k_BT)} \end{aligned}$$

using the geometric series. The single-mode free energy $F_\omega = -k_BT\ln Z_\omega = k_BT\ln(2\sinh(\hbar\omega/2k_BT))$. Summing over all $3N_{\mathrm{SC}}$ independent modes:

$$\boxed{F_{\mathrm{harm}}^{\mathrm{SOTC}}(T) = k_BT\sum_{\mathbf{q},s}\ln\left(2\sinh\frac{\hbar\omega^*_{\mathbf{q}s}}{2k_BT}\right)}$$

Thermodynamic quantities. Differentiating:

$$\begin{aligned} S_{\mathrm{harm}}(T) = k_B\sum_{\mathbf{q},s}\Bigg[&\frac{\hbar\omega^*_{\mathbf{q}s}}{2k_BT}\coth\left(\frac{\hbar\omega^*_{\mathbf{q}s}}{2k_BT}\right) \\ &- \ln\left(2\sinh\frac{\hbar\omega^*_{\mathbf{q}s}}{2k_BT}\right)\Bigg]. \end{aligned}$$

$$C_V^{\mathrm{harm}}(T) = k_B\sum_{\mathbf{q},s}\left(\frac{\hbar\omega^*_{\mathbf{q}s}}{2k_BT}\right)^2\frac{1}{\sinh^2\left(\hbar\omega^*_{\mathbf{q}s}/2k_BT\right)}$$

$$\mathrm{ZPE} = F_{\mathrm{harm}}(0) = \sum_{\mathbf{q},s}\frac{\hbar\omega^*_{\mathbf{q}s}}{2}$$

For a monatomic cubic crystal, the one-component displacement convention used in the main text is

$$U_{\mathrm{iso}}(T) = \frac{\hbar}{6N_cM}\sum_{\mathbf{q},s}\frac{1}{\omega^*_{\mathbf{q}s}}\coth\left(\frac{\hbar\omega^*_{\mathbf{q}s}}{2k_BT}\right).$$

Compounds require the species-resolved eigenvector-weighted form.

Limiting behavior. At $T \to 0$: $2\sinh(x) \to e^x$, so $F \to \sum\hbar\omega/2$ (ZPE). At $k_BT \gg \hbar\omega$: $\sinh(x) \to x$, giving $F \to k_BT\sum\ln(\hbar\omega_{\mathbf{q}s}/k_BT)$ (classical Dulong-Petit). The heat capacity $C_V \to 3N_{\mathrm{SC}}k_B$ at high T, correctly recovering the Dulong-Petit law.

Appendix I: Higher-Order Anharmonic Terms: Scope

FC2 alone cannot supply anharmonic free-energy corrections or three-phonon lifetimes; those quantities require explicit higher-order calculations. Wick's theorem for a chosen Gaussian trial distribution is a standard identity; variational free-energy minimization and cubic or quartic force constants are separate quantities. The validated quantities here are force-regression errors and fitted IFC tensors rather than higher-order free energies, linewidths, lifetimes, or thermal conductivity.

Appendix J: Implementation, Reproducibility, and Data Availability

Electronic-structure inputs. The exact SSSP filenames used by the archived QE calculations were `Ag_ONCV_PBE-1.0`, `Au_ONCV_PBE-1.0`, `Mo_ONCV_PBE-1.0`, `Al.pbe-n-kjpaw_psl.1.0.0`, `Cu.paw.z_11.ld1.psl.v1.0.0-low`, `Mg.pbe-n-kjpaw_psl.0.3.0`, `O.pbe-n-kjpaw_psl.0.1`, `Pb.pbe-dn-kjpaw_psl.0.2.2`, `cl_pbe_v1.4.uspp.F`, `na_pbe_v1.5.uspp.F`, `Te_pbe_v1.uspp.F`, and `W_pbe_v1.2.uspp.F`. The QE metallic-smearing input used `degauss=0.02` Ry. The Python interface flag `symmetrize_bonds=True` corresponds to the central-force projection used in the primary benchmarks.

Higher-order workflow and files. The independent FC2/FC3/FC4 fits use the `symfc` joint dense solver. The archive contains compact NumPy and HDF5 tensors, phono3py `fc3.hdf5`, ShengBTE `FORCE_CONSTANTS_3RD`, and FourPhonon `FORCE_CONSTANTS_4TH`. Synthetic tests check tensor shapes, index conventions, acoustic-sum-rule residuals, and cross-format consistency. The machine-readable campaign summaries store the parameter counts, normalized ranks, condition numbers, force-call accounting, and held-out metrics reported in Table III and Fig. 6.

The table is generated from the archived JSON, NPZ, and HDF5 data by `codes/validation/summarize_real_anharmonic_ifcs.py`.

The direct dispersion, canonical-subset, corrected-MSD, and reference-force audits are generated by `codes/validation/analyze_reviewer_evidence.py` and stored in `validation_results/reviewer_evidence_summary.json`; the independent NaCl non-analytic arrays and Fig. 4 are generated by `codes/validation/plot_nacl_loto.py`. Same-volume MgO and PbTe tensors are stored under `validation_results/mgo/dfpt_nac_production_volume` and `validation_results/pbte/dfpt_nac_production_volume`, generated by `codes/validation/run_dfpt_nac.py`. The selected-production polar arrays and Fig. 5 are generated by `codes/validation/plot_sotc_polar_nac.py` and summarized in `validation_results/production_sotc_nac_summary.json`. The matched sampling campaigns are run by `codes/validation/run_matched_sampling_ablation.py`; restartable QE states are checkpointed locally, while the public archive contains the compact material summaries used by `codes/validation/plot_matched_sampling_ablation.py`. All reported values are read directly from the machine-readable outputs.

Data and software availability. The open-source Python implementation, QE inputs, benchmark runners, fitted IFCs, compact validation summaries, selected plot arrays, and generated figures are available at `https://github.com/albert-hzbn/SOTC`. Machine-readable public results use JSON, NPZ, and HDF5 formats. Large QE scratch and restart directories are excluded from Git but can be regenerated from the archived inputs and drivers; the repository README states this boundary explicitly. Quantitative validation uses comparative arrays with calculation-level provenance. Quantum ESPRESSO [16] is available at `https://www.quantum-espresso.org`; the pseudopotentials are from the SSSP library [18], available at `https://www.materialscloud.org/discover/sssp`.

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